

Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays

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Abstract

A common anonymity pattern in overlay routing is *random-walk delegation*: an initiator performs a short random walk to pick a *delegate* and asks the delegate to execute a normal lookup. We develop a compact, proof-oriented anonymity theory for delegate-only observers.

Our organizing lens is structural: under a stationary initiator prior, the posterior distribution of the initiator given the observed delegate is exactly the t -step law of the *time-reversed* chain started at that delegate. This reduction turns delegate-only anonymity questions into standard mixing questions (including on directed/nonreversible overlays). We derive sharp distinguishing bounds in total variation distance and give leakage certificates: a conservative σ^t bound via $\ell_2(\pi)$ contraction and an *exact* expected χ^2 (Rényi-2) leakage identity in terms of the *full* singular spectrum of the discriminant operator.

For space, this manuscript includes a *reconstruction specification* and a small set of *sanity-check tables* rather than embedding full code and logs. A complete artifact bundle (scripts, raw outputs, and plotting) is available as supplementary material.

1 Introduction

Many privacy-preserving overlay protocols hide who initiated a lookup by forwarding the request through one or more intermediate relays, often selected by random walks. Random-walk selection appears across anonymous DHT lookup and relay-discovery systems (e.g., Octopus [8], ShadowWalker [6], NISAN [7]) and in early foundational models such as Crowds.

Primitive (fixed-length delegation). An initiator S runs a length- t random walk on the overlay to sample a delegate D , then asks D to execute a standard lookup. Our core threat model assumes the adversary learns only the delegate identity D for a single delegation. (We briefly discuss “first-hit” compromise observations in section 10.)

What is new here. The posterior-as-time-reversal identity is elementary, but it is *useful as a reduction*: it makes the anonymity analysis of delegation *exactly* a mixing analysis problem, including for directed/nonreversible overlays. The main technical contributions are the resulting optimal distinguishability bounds and computable leakage certificates for directed chains, including a full-spectrum (not only σ_2) identity.

Main results. Let P be the walk kernel on node set V with stationary π and let $P^\star$ be its time reversal.

- **Posterior structure (tool).** Under $S \sim \pi$, the posterior is a reversed walk: $\mathbb{P}[S = \cdot \mid D = d] = (P^\star)^t(d, \cdot)$ (theorem 5.1).
- **Optimal tests.** Any test distinguishing two candidate initiators from the delegate identity has optimal advantage equal to total variation distance between their delegate laws (corollary 5.5).
- **Spectral certificates, including full spectrum.** Let $\mathcal{D} = \Pi^{1/2} P \Pi^{-1/2}$ be the discriminant operator with singular values $1 = \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$. We give (i) a conservative bound $\mathbb{E}[\chi^2(\Pr[S = \cdot \mid D] \parallel \pi)] \leq \sigma_2^{2t}$ and (ii) an exact identity $\mathbb{E}[\chi^2(\Pr[S = \cdot \mid D] \parallel \pi)] = \sum_{i=2}^n \sigma_i^{2t}$ (theorem 6.2).
- **Perfect delegation.** Stopping at a strong stationary time yields a delegate distributed exactly as π and independent of S (perfect anonymity against delegate-only observers; section 7).
- **Non-normal cautionary example.** Directed de Bruijn overlays mix in finite time yet can have $\sigma_2 \approx 1$, making σ_2^t bounds vacuous (section 11).

Limitations (explicit). This paper deliberately targets *delegate-only* observation and a *fixed* kernel P . We do *not* model global passive traffic analysis (timing/volume correlation, batching, packet sizes), active/adaptive adversaries (e.g., dropping/biasing forwarding, Sybil/route-capture), or multi-point observation beyond the delegate identity. We also do not model churn and other time-variation in the overlay itself: if the kernel changes over time (P_t), the certificates in this paper should be read as snapshot guarantees and can degrade toward a noise floor as churn “resets” the mixing clock.

Artifacts and reconstruction. To keep the manuscript compact, we include only a reconstruction specification and sanity-check tables. Full scripts, raw outputs, and plotting are available as supplementary material.

2 How to read and verify (LLM-friendly)

This paper is organized around two “hinge” facts that let you translate delegation anonymity questions into standard Markov-chain objects.

- **Hinge 1 (posterior as time reversal).** Given a delegate $D = d$ under a stationary initiator prior, the posterior on the initiator is *exactly* a reversed walk started at d (theorem 5.1). You can treat this as a reduction: any bound on how fast a chain “forgets its start” can be imported as an anonymity statement for delegation.
- **Hinge 2 (expected χ^2 leakage as a spectrum).** The expected Rényi-2 leakage has an *exact* formula in terms of the singular spectrum of the discriminant operator (theorem 6.2). This supplies a computable certificate that remains informative for directed/non-normal overlays where the single-number σ_2 bound can be misleading.

Fast verification path. A quick referee-style check is: (i) verify theorem 5.1 by Bayes + time reversal; (ii) verify theorem 6.2 by rewriting expected collision probability as a Frobenius norm; (iii) reproduce tables 3 and 5 from the reconstruction spec in section 14. Full scripts and raw outputs are available as supplementary material.

3 Related work and context

Random-walk-based relay selection has a long history in anonymity systems, from early models such as *Crowds* to anonymous lookup schemes for structured overlays. Our focus is deliberately narrow: we isolate the *delegation* primitive and ask what can be inferred from the *delegate identity alone*. This lens complements system papers that treat richer adversaries (timing, volume, route-capture, Sybils) by providing sharp and composable bounds for the delegate-only component.

For concrete systems that instantiate random-walk relay selection and/or anonymous lookup in structured overlays, see ShadowWalker [6] (random-walk-based peer-to-peer anonymity over structured topologies), NISAN [7] (DHT-based network information service for anonymization networks), and Octopus [8] (secure and anonymous DHT lookup).

On the Markov-chain side, we use standard notions from mixing theory—total variation and separation distance, time reversal, and strong stationary times. Background can be found in Levin–Peres–Wilmer [1] and in the original strong-stationary-time work of Aldous–Diaconis [2]; see also Fill [3] for related perspectives. On the systems side, our results provide a principled way to interpret (and sometimes refute) common heuristics that “a spectral gap guarantees privacy,” especially for directed overlays. We also relate to work on adding query privacy to robust DHTs [5] and to classic relay-anonymity models [4].

4 Model and definitions

Let V be a finite set of nodes, $|V| = n$. Let P be a (row-stochastic) Markov kernel on V with stationary distribution π satisfying $\pi(x) > 0$ for all x . We allow P to be directed and nonreversible.

Delegation experiment. Sample an initiator $S \sim \pi$. Starting from S , run the chain for t steps and output the endpoint D (the delegate):

$$\mathbb{P}[D = d \mid S = s] = P^t(s, d).$$

Time reversal. Define the time-reversed kernel P^* by

$$\pi(x)P(x, y) = \pi(y)P^*(y, x) \quad (\forall x, y \in V), \tag{1}$$

so $P^*(y, x) = \pi(x)P(x, y)/\pi(y)$.

Distances and leakage. For distributions μ, ν on V ,

$$\text{TV}(\mu, \nu) = \frac{1}{2} \sum_{x \in V} |\mu(x) - \nu(x)|.$$

The Pearson chi-squared divergence is

$$\chi^2(\mu \parallel \pi) = \sum_{x \in V} \frac{(\mu(x) - \pi(x))^2}{\pi(x)} = \sum_{x \in V} \frac{\mu(x)^2}{\pi(x)} - 1.$$

We use conditional min-entropy $H_\infty(S \mid D) = -\log \sum_d \mathbb{P}[D = d] \max_s \mathbb{P}[S = s \mid D = d]$.

5 Posterior structure and optimal tests

Theorem 5.1 (Posterior equals reversed walk). *Assume $S \sim \pi$ and D is obtained by a length- t walk from S using P . Then for every $d \in V$,*

$$\mathbb{P}[S = s \mid D = d] = (P^\star)^t(d, s) \quad (\forall s \in V).$$

Proof. Bayes' rule gives $\mathbb{P}[S = s \mid D = d] = \frac{P^t(s, d)\pi(s)}{\sum_u P^t(u, d)\pi(u)}$. Stationarity implies $\sum_u \pi(u)P^t(u, d) = \pi(d)$, hence $\mathbb{P}[S = s \mid D = d] = \pi(s)P^t(s, d)/\pi(d)$. Applying (1) t times yields $\pi(s)P^t(s, d) = \pi(d)(P^\star)^t(d, s)$. $\square$

Proposition 5.2 (Time-varying kernels). *The posterior-as-reversal lens extends verbatim to time-inhomogeneous walks. Let $P_1, \dots, P_t$ be a sequence of Markov kernels that share a stationary distribution π (i.e., $\pi P_i = \pi$ for all i). Sample $S \sim \pi$ and run the inhomogeneous chain $X_0 = S$, $X_i = X_{i-1}P_i$, and output $D = X_t$. Then, for every $d \in V$,*

$$\mathbb{P}[S = s \mid D = d] = (P_t^\star P_{t-1}^\star \cdots P_1^\star)(d, s),$$

where $P_i^\star$ is the time reversal of P_i with respect to π .

Proof. By Bayes' rule, $\mathbb{P}[S = s \mid D = d] = \pi(s)(P_1 \cdots P_t)(s, d)/\pi(d)$, using $\mathbb{P}[D = d] = \pi(d)$. Applying the single-step reversal identity $\pi(x)P_i(x, y) = \pi(y)P_i^\star(y, x)$ along the product yields $(P_t^\star \cdots P_1^\star)(d, s)$. $\square$

Remark 5.3 (Churn as perturbation). *If churn changes the kernel over time but approximately preserves a common stationary distribution, proposition 5.2 shows that the anonymity analysis is still an exact reversal of the actual time-varying walk. If stationarity itself drifts, the snapshot results in this paper should be interpreted as local guarantees that can degrade toward a noise floor.*

Remark 5.4 (Framing). *theorem 5.1 is a one-line calculation, but it is the key lens of this paper: it converts delegate-only anonymity analysis into standard mixing analysis on the reversed chain, including for directed/nonreversible overlays.*

Corollary 5.5 (Optimal test). *For any two candidates $s_0, s_1 \in V$ and any (possibly randomized) test $\phi : V \rightarrow \{0, 1\}$ based only on D ,*

$$\max_{\phi} \left| \mathbb{P}[\phi(D) = 1 \mid S = s_0] - \mathbb{P}[\phi(D) = 1 \mid S = s_1] \right| = \text{TV}(P^t(s_0, \cdot), P^t(s_1, \cdot)).$$

Proof. Classical characterization of total variation distance as optimal distinguishing advantage between two distributions. $\square$

6 Spectral anonymity bounds

For reversible chains, mixing is governed by eigenvalues; for directed chains, singular values of the *discriminant* operator play the analogous role.

Definition 6.1 (Discriminant operator). *Let $\Pi = \text{diag}(\pi)$. Define*

$$\mathcal{D} = \Pi^{1/2} P \Pi^{-1/2}.$$

Let $1 = \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ be its singular values.

Theorem 6.2 (Exact expected chi-squared leakage). *Under the stationary prior, let $\mu_d(\cdot) = \mathbb{P}[S = \cdot \mid D = d]$. Then*

$$\mathbb{E}_{D \sim \pi} [\chi^2(\mu_D \parallel \pi)] = \sum_{i=2}^n \sigma_i^{2t}.$$

Proof sketch. Let $M = \Pi^{1/2} P^t \Pi^{-1/2} = \mathcal{D}^t$. One can rewrite $\mathbb{E}_{D \sim \pi} [\sum_s \mu_D(s)^2 / \pi(s)]$ as the Frobenius norm $\|M\|_F^2$, which equals the sum of squared singular values of M , namely $\sum_i \sigma_i^{2t}$. Subtract the $i = 1$ term. $\square$

Corollary 6.3 (Conservative σ_2 certificate). *Since $\sum_{i=2}^n \sigma_i^{2t} \leq (n-1)\sigma_2^{2t}$,*

$$\mathbb{E}_{D \sim \pi} [\chi^2(\mu_D \parallel \pi)] \leq (n-1)\sigma_2^{2t}, \quad \mathbb{E}[\text{TV}(\mu_D, \pi)] \leq \frac{1}{2} \sqrt{(n-1)} \sigma_2^t.$$

Why full spectrum matters. For non-normal directed chains, σ_2 can be close to 1 even when the chain mixes quickly (see section 11); theorem 6.2 can remain sharp because most singular values may be small or zero.

7 Perfect delegation via strong stationary times

Definition 7.1 (Strong stationary time). *A stopping time T is a strong stationary time if $X_T \sim \pi$ and X_T is independent of T .*

Theorem 7.2 (Perfect anonymity against delegate-only observers). *If delegation outputs $D = X_T$ for a strong stationary time T , then $D \sim \pi$ and D is independent of S . Hence $\mathbb{P}[S = s \mid D = d] = \pi(s)$ for all s, d .*

Proof. Immediate from the definition of strong stationary time and independence. $\square$

8 Repeated lookups: tensorization

Suppose the same initiator performs k independent delegations, producing delegates $D_1, \dots, D_k$.

Proposition 8.1 (Posterior tensorizes). *Under a stationary prior and conditional independence of walks,*

$$\mathbb{P}[S = s \mid D_1 = d_1, \dots, D_k = d_k] \propto \pi(s) \prod_{j=1}^k P^t(s, d_j).$$

Proof. Bayes' rule and conditional independence: $\mathbb{P}[D_1 = d_1, \dots, D_k = d_k \mid S = s] = \prod_j P^t(s, d_j)$. $\square$

Amplification (interpretation). The optimal distinguishing advantage between two candidates from the vector $(D_1, \dots, D_k)$ is the total variation distance between the corresponding product laws $P^t(s_i, \cdot)^{\otimes k}$. For large n , exact computation of product-TV or Chernoff information is typically impractical; in practice, one can upper-bound amplification using additive divergences (e.g., KL) or use the single-delegation certificates as conservative proxies.

9 Randomized walk length and hopcount leakage

Some systems randomize the walk length (hopcount) to reduce latency or to complicate inference. Let T be an integer-valued random walk length, sampled independently of the initiator and of the walk transitions, and let the delegate be $D = X_T$.

Proposition 9.1 (Posterior under randomized length). *Assume $S \sim \pi$ and $T \perp S$. Then:*

1. *If the adversary observes both $(D, T) = (d, t)$, then $\mathbb{P}[S = \cdot \mid D = d, T = t] = (P^*)^t(d, \cdot)$.*
2. *If the adversary observes only $D = d$, then the posterior is a mixture of reversed walks:*

$$\mathbb{P}[S = s \mid D = d] = \sum_{t \geq 0} \mathbb{P}[T = t] (P^*)^t(d, s).$$

Proof. For (1), condition on $T = t$ and apply theorem 5.1. For (2), $\mathbb{P}[S = s \mid D = d] \propto \pi(s) \sum_t \mathbb{P}[T = t] P^t(s, d)$ and the denominator equals $\pi(d)$ by stationarity; rewrite each term using time reversal. $\square$

Design guidance. Randomizing length only helps if it prevents the adversary from conditioning on short walks. In particular, if T has nontrivial probability on small values, then the mixture in proposition 9.1 retains a “short-walk” component that can dominate leakage. A practical pattern is *burn-in plus a tail*: choose a minimum $t_{\min}$ that already meets a leakage target, and then add a random tail (e.g., geometric) whose realization is hidden by padding or batching. Full analyses for specific hopcount distributions are included in the artifact bundle.

10 Compromised nodes and “early observation”

A common deviation from the pure delegate-only model is that an adversary who compromises a set $C \subseteq V$ may learn the first time a walk hits C (and possibly which compromised node was hit first). Even without timing, this “first-hit” observation can leak information beyond the delegate identity.

We treat a simple time-hidden model in section 13 as a robustness check: the adversary learns only the identity of the first compromised node hit within a fixed horizon, if any.

11 Exact mixing on directed de Bruijn overlays

The directed de Bruijn graph $B(k, m)$ has $n = k^m$ nodes labeled by length- m strings. A move shifts left and appends a fresh symbol. The non-lazy walk mixes *exactly* in m steps (the last m appended symbols determine the state), yet the walk is highly non-normal.

Remark 11.1 (Finite-time mixing vs σ_2). *On the non-lazy de Bruijn walk, the usual σ_2^t intuition can mislead: the chain reaches stationarity in m steps, but singular-value bounds can be vacuous if $\sigma_2 \approx 1$. Full-spectrum leakage identities are therefore a better diagnostic on directed overlays.*

Deterministic suffix leakage. In the non-lazy de Bruijn walk, the delegate reveals a deterministic suffix of the initiator (the last $m - t$ symbols) because shifting discards a prefix. This illustrates that “mixing in distribution” and “hiding structured attributes” can diverge unless the threat model and protocol details are aligned.

12 Practical guidance: which bound to use

This section summarizes how to turn the theory into “engineering numbers” and how to match a leakage question to a certificate.

Goal	Suggested quantity	How to estimate / interpret
Best test between two candidate initiators	$\text{TV}(P^t(s_0, \cdot), P^t(s_1, \cdot))$	Exact optimal advantage (corollary 5.5); estimate by Monte Carlo when n is large.
Average “distance from uniform” of the posterior	$\mathbb{E}[\chi^2(\Pr[S = \cdot \mid D] \parallel \pi)]$	Exact identity $\sum_{i \geq 2} \sigma_i^{2t}$ (theorem 6.2); approximate using a few leading singular values.
Conservative quick bound on average leakage	σ_2^{2t}	Easy to compute but can be loose on non-normal digraphs; use as a pessimistic upper bound (corollary 6.3).
Worst-case spikes	separation distance	Controls the maximum posterior ratio; estimate via $\min_x P^t(s, x)/\pi(x)$ (simulation or sparse powering).

Table 1: Certificate menu for delegate-only leakage questions.

Threat-model ladder (what the adversary observes). The delegate-only model is the baseline. Conditioning on additional side information typically strengthens the adversary; the posterior lens remains the right way to incorporate it:

- **Delegate only:** observe D ; posterior is a reversed walk (theorem 5.1).
- **Delegate + hopcount:** observe (D, T) ; posterior is $(P^*)^T(d, \cdot)$ (proposition 9.1).
- **First-hit compromise:** observe first compromised node hit (possibly time-hidden); see section 10.
- **Global timing/volume, route capture, Sybils:** out of scope (see section 1).

Design recommendations (turning bounds into protocol knobs)

This subsection translates the theory into actionable choices of walk kernel, walk length, and operational mitigations.

R1. Pick an attacker metric and set a numeric target. If the adversary is comparing two candidates, total variation is the right unit: target ε and choose parameters so that $\text{TV}(P^t(s_0, \cdot), P^t(s_1, \cdot)) \leq \varepsilon$ for the relevant pairs (corollary 5.5). If you want “posterior close to prior on average,” target $\mathbb{E}[\chi^2(\Pr[S = \cdot \mid D] \parallel \pi)] \leq \delta$ or $\mathbb{E}[\text{TV}(\Pr[S = \cdot \mid D], \pi)] \leq \varepsilon$.

R2. Use the full-spectrum certificate when P is directed. On directed/non-normal overlays, σ_2 can dramatically mis-predict privacy (section 11). When possible, estimate the leading singular values of the discriminant operator and choose the smallest t such that $\sum_{i \geq 2} \sigma_i^{2t} \leq \delta$ (theorem 6.2). If you only have σ_2 , treat σ_2^{2t} as a conservative upper bound (corollary 6.3), not a calibrated estimate.

R3. Fix structural leakage in the kernel before tuning t . A longer walk does not help if the kernel deterministically preserves identifying structure (e.g., a shift-like update on labels, section 11). Low-cost kernel modifications that usually improve privacy are: (i) add laziness $P_{\text{lazy}} = \alpha I + (1 - \alpha)P$, (ii) add a small teleport mixture $\alpha \mathbf{1}\pi^\top + (1 - \alpha)P$, and (iii) prefer Eulerian/near-regular overlays so π is simple.

R4. Make repeated queries a first-class design dimension. If an observer can link k delegations to the same initiator, posteriors multiply (section 8) and leakage can amplify. Either increase t to meet targets at the session level, or break linkage via key rotation/pseudonyms, rate limits, batching, or enforced cool-down windows.

R5. If you randomize hopcount, hide it and include burn-in. Revealed hopcount strengthens the adversary: conditioning on T produces a sharper posterior (section 9). If you randomize length, keep hopcount hidden and use a *burn-in plus tail* distribution (e.g., $T = t_0 + G$ with geometric G) so every delegation mixes at least t_0 steps while the tail obscures exact length.

R6. Treat churn as a noise floor; re-estimate periodically. The clean certificates assume a fixed P . With churn, the operational kernel is time-varying. If instantaneous kernels share π , the reversal lens extends to inhomogeneous products (proposition 5.2); otherwise churn perturbs both P and π , typically imposing a residual leakage floor. In practice: add modest laziness/teleportation, avoid brittle deterministic routing steps, and periodically recompute certificates on current snapshots.

R7. Prefer strong stationary times when feasible. An SST-based stop yields perfect delegate-only anonymity (section 7). If you can implement an exact or approximate SST, it dominates fixed- t tuning.

Implementation recipe. (1) choose the threat-model rung; (2) choose a leakage unit (TV or expected χ^2) and a numeric target; (3) choose a privacy-friendly kernel; (4) estimate a certificate (prefer full-spectrum); (5) pick the smallest t meeting the target; (6) revisit if linkage or churn changes your regime.

Design recommendations (expanded: protocol knobs, calibration, and pitfalls)

The menu in table 1 tells you *which* certificate matches a question. This section tells you *what to do* with those numbers.

Knobs-to-effects.

Calibration patterns (how to pick t). Two simple patterns cover many deployments.

- **Pairwise indistinguishability target (TV).** Identify the pairs of starts that matter (e.g., any two initiators in the same ASN/rack/region, or any two users in a cohort). Choose t so that $\max_{(s_0, s_1) \in \mathcal{P}} \text{TV}(P^t(s_0, \cdot), P^t(s_1, \cdot)) \leq \varepsilon$. This aligns directly with optimal tests (corollary 5.5). At large n , estimate each TV by sampling delegates from each start and using a plug-in TV estimate or a held-out classifier.
- **Average posterior drift target (expected χ^2).** Choose δ and pick the smallest t with $\sum_{i \geq 2} \sigma_i^{2t} \leq \delta$ (theorem 6.2). When computing the full spectrum is too costly, estimate a few leading σ_i and bound the remainder by $(n - r)\sigma_{r+1}^{2t}$.

Knob	What it buys (privacy)	Cost / how to validate
Increase walk length t	Reduces delegate-only leakage as the reversed walk mixes; the sharpest statement is the optimal TV test (corollary 5.5) and the expected χ^2 identity (theorem 6.2).	Latency, bandwidth, and exposure to compromise during the walk. Validate with pairwise TV on the cohort you care about, and/or $\sum_{i \geq 2} \sigma_i^{2t}$.
Add laziness / self-loops	Damps periodicity, smooths worst-case spikes, and makes metrics more stable under mild nonstationarity.	Adds hops. Validate by re-estimating $t_{0.1}$ and checking separation distance.
Small teleport mixture to π (or an approximate directory)	Creates a robust “mixing floor” even under churn; combats deterministic structure and non-normal transients.	Requires a sampling primitive approximating π ; can be amortized via cached samples. Validate by checking that the teleport mass dominates the minimum-probability ratio.
Hide hopcount (and use burn-in + tail)	Prevents conditioning on T ; the posterior becomes a mixture of reversed walks (proposition 9.1).	Must avoid side channels that leak length (timeouts, packet counts, retransmits). Validate by evaluating both (D-only) and (D,T) attack models.
Break linkability across queries	Prevents tensorization amplification (section 8); makes “per-query” tuning meaningful.	Requires protocol-level unlinkability (ephemeral IDs, batching, rate limits). Validate by enumerating what an adversary can realistically link.
SST / near-SST stopping	If exact SST is implementable, yields perfect delegate-only anonymity (section 7) and makes stopping adaptive.	Harder to implement; may require additional randomness or certificates. Validate by checking that delegates are empirically π -distributed for many starts.

Table 2: Protocol knobs and what the mathematics predicts they do.

Estimating certificates when n is huge. The theory is finite- n , but estimation is often sublinear. Useful options:

- **Sampling-based TV/separation.** If you can run many delegations from a start s , estimate $P^t(s, \cdot)$ empirically and compute TV/separation against a baseline. This is the most system-faithful approach.
- **Randomized SVD for top singular values.** On sparse kernels, randomized methods estimate the top σ_i of the discriminant operator $\mathcal{D}$ using only matrix–vector products.
- **Stochastic trace estimation for $\sum \sigma_i^{2t}$.** Since $\sum_i \sigma_i^{2t} = \|\mathcal{D}^t\|_F^2 = \text{tr}((\mathcal{D}^\top \mathcal{D})^t)$, one can estimate $\text{tr}(A^t)$ for $A = \mathcal{D}^\top \mathcal{D}$ with Hutchinson estimators ($z^\top A^t z$ for random Rademacher z) using repeated sparse multiplies by A . This targets the expected- χ^2 certificate directly.

Common pitfalls.

- **Mixing-to- π from one start is not enough.** You want indistinguishability between *different* starts. Check pairwise TV (or a max-over-cohort) rather than only $\text{TV}(P^t(s, \cdot), \pi)$.
- **Non-normality breaks “ σ_2 intuition.”** On directed overlays treat σ_2 as a conservative bound, not as a calibrated prediction (section 11).

- **Side channels can reintroduce hopcount.** If length can be inferred from timing, packet counts, or failure behavior, then the relevant posterior is conditioned on T (section 9).
- **Churn can dominate.** If the overlay changes substantially over the delegation window, certificates computed on stale snapshots may be irrelevant. Treat churn as a background “leakage floor” and re-estimate on rolling snapshots (section 1).

Speculative but plausible: continuous privacy control loops. Because building evaluators is now cheap, it is plausible to operationalize the mathematics as a control loop: snapshot the overlay into a sparse kernel P ; estimate certificates for a grid of t ; pick the smallest t meeting policy; and alert when churn or compromise pushes measured leakage above threshold. Even if estimates are noisy, this replaces folklore with measurable guardrails.

13 Sanity-check experiments (compact)

We include two small tables to make reconstruction easy and to anchor the scale of the bounds. Full experiments, additional sweeps, and plotting are available as supplementary material.

Family comparison at $n = 1024$. We compare 1/2-lazy random-walk mixing behavior on several $n = 1024$ -node regular graph families, including a simple Chord-like ring-with-fingers digraph. For each family we report the measured time $t_{0.1}$ to reach $\text{TV}(P^t(0, \cdot), \pi) \leq 0.1$, along with separation distance at that time. We also report a conservative σ -based prediction obtained by solving $\frac{1}{2}\sqrt{n}\sigma^t \leq 0.1$ (using $\ell_2 \rightarrow \ell_1$).

Family	d	σ	measured $t_{0.1}$	TV@ $t_{0.1}$	Sep@ $t_{0.1}$	pred. $t_{0.1}$
Cycle C_n	2	0.999991	196657	0.1000	0.3129	539201
2D torus (32×32)	4	0.995196	438	0.1000	0.4264	1054
Hypercube Q_{10}	10	0.900000	25	0.0904	0.5634	49
Chord ring+fingers	10	0.900000	18	0.0989	0.7633	49
Random 8-regular	8	0.826246	15	0.0905	0.3325	27
Directed de Bruijn $B(2, 10)$	2	0.988831	24	0.0769	0.1537	452

Table 3: Sanity-check family comparison ($n = 1024$, 1/2-lazy walks, fixed start 0).

Visualization: smooth vs. cliff mixing. To make the non-normal cautionary point visual, figure 1 plots $\text{TV}(P^t(0, \cdot), \pi)$ for a 1/2-lazy cycle on $n = 1024$ (slow, smooth decay) against the *non-lazy* directed de Bruijn walk on $B(2, 10)$ (exact mixing with a sharp cutoff at $t = 10$). This figure is illustrative; the tables report the 1/2-lazy kernels used throughout.

Scaling check for the Chord ring+fingers overlay. Because the ring+fingers kernel is a circulant (Cayley) digraph on $\mathbb{Z}_n$, the distribution and $\sigma = \max_{k \neq 0} |\lambda_k|$ can be computed exactly by FFT. table 4 reports $t_{0.1}$ and separation as $n = 2^m$ grows.

Sensitivity to compromise fraction: time-hidden first-hit. We sweep compromised fraction α and report (i) the probability of hitting the compromised set within a horizon of 30 steps and (ii) leakage in a *time-hidden* model where the adversary learns only which compromised node was hit first. (Construction details are in section 14.)

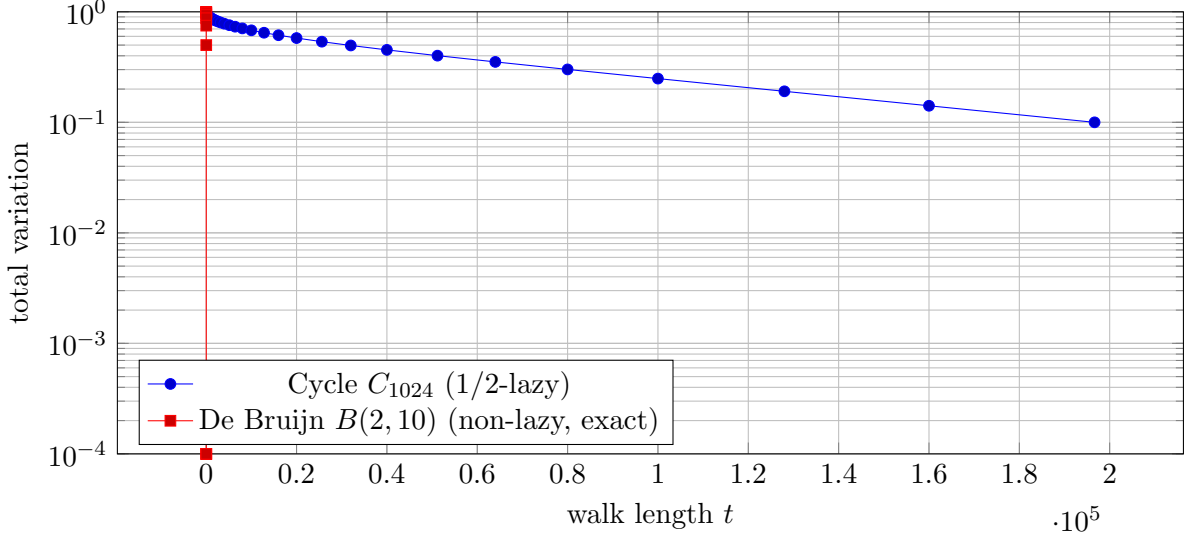


Figure 1: Smooth decay vs. finite-time cutoff in TV. For log- y display, exact zeros are clipped to 10^{-4} .

n	m	σ	measured $t_{0.1}$	TV@ $t_{0.1}$	Sep@ $t_{0.1}$	pred. $t_{0.1}$
256	8	0.875000	14	0.0963	0.7028	33
512	9	0.888889	16	0.0978	0.7346	41
1024	10	0.900000	18	0.0989	0.7633	49
2048	11	0.909091	21	0.0876	0.7270	57

Table 4: Chord ring+fingers scaling (1/2-lazy, start 0).

14 Reconstruction specification (compact)

This section specifies enough to reconstruct tables 3 and 5 from scratch. The goal is *interoperable reimplementations*: an independent implementation (human or LLM-generated) should reproduce values up to small numerical tolerance.

14.1 Graph families and kernels

All kernels use **1/2-laziness**: with probability 1/2 stay put; with probability 1/2 take a uniform random outgoing step. All graphs are on $V = \{0, 1, \dots, n-1\}$ with $n = 1024$ unless stated.

- **Cycle C_n .** Undirected edges $(i, i \pm 1 \bmod n)$. Lazy walk as above.
- **2D torus.** Grid $(\mathbb{Z}/32\mathbb{Z})^2$ mapped to 0..1023; neighbors differ by ± 1 in one coordinate (4-neighborhood); lazy walk.
- **Hypercube Q_{10} .** Nodes are 10-bit strings; neighbors differ in one bit (degree 10); lazy walk.
- **Random 8-regular.** Sample an undirected 8-regular graph uniformly at random (e.g., `networkx.random_regular_graph`) with RNG seed 0; lazy walk.
- **Chord ring+fingers (directed).** Let $n = 2^{10}$. From node x , out-neighbors are $x + 2^j \bmod n$ for $j = 0, \dots, 9$ (outdegree 10). Use the lazy out-neighbor walk. (This digraph is regular and

comp. frac	Chord ring+fingers			Random 8-regular		
	$P[\text{hit} \leq 30]$	I_{hid} (bits)	MAP_{hid}	$P[\text{hit} \leq 30]$	I_{hid} (bits)	MAP_{hid}
0.01	0.135986	0.046782	0.002159	0.118898	0.041077	0.002449
0.05	0.526431	0.263719	0.007143	0.486559	0.237269	0.008565
0.10	0.779561	0.587120	0.013865	0.742507	0.534519	0.017125
0.20	0.958342	1.365357	0.029827	0.930654	1.236780	0.036783

Table 5: Sanity-check “first-hit” sensitivity to compromised fraction ($n = 1024$, horizon 30).

Eulerian, so π is uniform.)

- **Directed de Bruijn** $B(2, 10)$. Nodes are length-10 bitstrings. A move shifts left and appends a fresh random bit; use 1/2-laziness. (Uniform π .)

14.2 Metrics and procedures

Stationary distribution. For the regular families above, $\pi = \text{Unif}(V)$. If implementing variants that are not Eulerian, compute π by power iteration until $\|\pi P - \pi\|_\infty$ is below 10^{-12} .

TV and separation. Starting from the point mass at node 0, iterate $p_{t+1} = p_t P$. Compute $\text{TV}(p_t, \pi) = \frac{1}{2} \sum_x |p_t(x) - \pi(x)|$ and $\text{sep}(p_t, \pi) = 1 - \min_x p_t(x)/\pi(x)$.

$t_{0.1}$. Define $t_{0.1} = \min\{t : \text{TV}(p_t, \pi) \leq 0.1\}$.

Singular value σ . Compute $\sigma = \sigma_2(\mathcal{D})$ where $\mathcal{D} = \Pi^{1/2} P \Pi^{-1/2}$. For uniform π , $\mathcal{D} = P$. In sparse form, estimate the top singular values via Lanczos/ARPACK SVD; report σ_2 to 6 decimals.

Predicted $t_{0.1}$. Solve for the smallest integer t such that $\frac{1}{2}\sqrt{n}\sigma^t \leq 0.1$ (using $\|v\|_1 \leq \sqrt{n}\|v\|_2$).

14.3 First-hit time-hidden compromise sweep

Fix horizon $H = 30$ and compromised fraction $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$. For each α , sample a compromised set C of size $\lfloor \alpha n \rfloor$ uniformly at random (seed 0). Sample $S \sim \text{Unif}(V)$ and run a length- H lazy walk from S . Let Y be the identity of the *first* node in C hit (if any), else a special symbol $\perp$. In the *time-hidden* model the adversary observes Y .

Estimate:

$$I_{\text{hid}} = I(S; Y), \quad \text{MAP}_{\text{hid}} = \mathbb{P}[\hat{S}(Y) = S] \text{ for the Bayes-optimal } \hat{S}.$$

Use Monte Carlo with at least 10^5 trials per α (the artifact uses more). Treat results within ± 0.02 bits (MI) and $\pm 2 \times 10^{-3}$ (MAP) as matching.

Sanity anchors. An implementation consistent with the above should reproduce tables 3 and 5 within the stated tolerances.

15 Conclusion

Time reversal makes random-walk delegation anonymity explicit: the observed delegate reveals a reversed walk. This yields sharp optimal distinguishability and computable leakage certificates for directed overlays, and clarifies why single-parameter spectral heuristics can fail on non-normal kernels. We provide compact reconstruction guidance here to enable straightforward reimplementations.

Artifact availability. A full artifact bundle (scripts, full parameter sweeps, and raw outputs used to generate all tables and figures) is intended to accompany this manuscript as supplementary material. To keep the main paper compact, we include only a reconstruction specification and a small set of sanity-check tables; an independent reimplementations should be straightforward from section 14.

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